SHAKIBUL CHOWDHURY

Astrophysics Researcher
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Research Summary

Researcher in relativistic astrophysics with interests in black holes, neutron stars, and observable signatures of strong gravity. My work focuses on polarization transport, relativistic timing phenomena, and radiation transport in Kerr spacetime, with applications to X-ray polarimetry (IXPE), black-hole spin diagnostics, and compact-object astrophysics. I develop computational frameworks for strong-gravity observables using numerical methods in Python and C++.

Education

City College of New York (CUNY)	M.S. Physics, 2025
Stony Brook University (SUNY)	B.S. Physics and Astronomy, 2020
Queensborough Community College (CUNY)	Physics and Mathematics, 2014–2016

Research Experience

Relativistic Polarization and Timing in Kerr Spacetime (2025–present)
Mentor: Prof. Joshua Tan (CUNY/AMNH)

- Developed a computational framework for general relativistic polarization transport in Kerr spacetime using stable observer-screen gauge conditions and RK4 geodesic integration.
- Published sole-author paper in *The Astrophysical Journal*: *Kerr Polarization Transport: Accuracy and Performance in General Relativistic Light Propagation* (2026).
- Investigating spin-dependent polarization signatures, relativistic timing asymmetries, full-Stokes imaging, and neutron-star polarization transport in strong-gravity environments.

Gaia-Based Photometric Analysis of HMXBs (2025)
Collaborator: Prof. Joshua Tan (CUNY/AMNH)

- Performed Gaia-calibrated photometric reduction and aperture photometry of the HMXB RX J0111.2–7317 using RGB observations and comparison-star calibration techniques.
- Measured post-outburst optical brightening using Gaia DR3 photometry and bootstrap uncertainty estimation.

Neutron-Star Equation of State and Tidal Deformability (2019–2020)
Mentor: Prof. James M. Lattimer (Stony Brook University)

- Computed neutron-star mass–radius relations and tidal deformabilities from tabulated equation-of-state models.
- Compared theoretical predictions with GW170817 constraints to study viable neutron-star EOS families.

Independent Study & Computational Astrophysics

- Completed an independent-study course in relativistic astrophysics and computational methods during the final semester of the M.S. Physics program at CCNY.
- Developed independent computational projects in Kerr spacetime polarization transport, relativistic timing asymmetries, and strong-gravity observables using Python and C++ numerical methods.
- Built backend C++ implementations for full-Stokes polarization imaging and relativistic ray-tracing workflows.

Teaching & Outreach

Teaching Assistant & Lab Instructor, CCNY Physics (2023–present)
Support PHYS 207/208 calculus-based laboratory courses; guide students through experimental methods, data analysis, uncertainty estimation, and scientific reporting. Provide instruction during weekly laboratory sessions and detailed feedback on lab reports and quantitative reasoning.

Teaching Assistant, Simply ED LLC (2023–present)
Assist instruction in Algebra 1 and Geometry courses for grades 9–12; support classroom problem-solving, reinforce core mathematical concepts, and provide individualized academic support to strengthen student understanding and performance.

Regents Prep Tutor, Truman High School (NYC) (2022–present)
Tutor Algebra 2 and Geometry for New York Regents preparation; develop customized practice materials, structured review plans, and problem-solving strategies tailored to individual student needs.

Student Support Services, Stony Brook University (2018–2020)
Provided academic-note support for students registered with Disability Support Services through

accurate and detailed lecture-note preparation across multiple STEM courses while maintaining student confidentiality.

Outreach Intern, AstroCamp (California) (2022)
Led astronomy outreach activities for K–12 students, including telescope observing sessions, night-sky tours, and hands-on demonstrations in astronomy and physics.

Science Communication Intern, Appalachian Mountain Club (2017)
Developed astronomy-focused educational materials and contributed to public science outreach programs for community and outdoor education events.

Publications

Chowdhury, S. (2026). *Kerr Polarization Transport: Accuracy and Performance in General Relativistic Light Propagation. The Astrophysical Journal.* DOI: 10.3847/1538-4357/ae6652.

Developed and validated a computational framework for general relativistic polarization transport in Kerr spacetime using stable observer-screen gauge conditions and efficient RK4 integration methods for modeling observable X-ray polarization signatures near black holes.

Submitted Manuscripts

Chowdhury, S. *A New Timing Signature of Black Hole Spin: Time-Delay Asymmetry in Kerr Spacetime*. Submitted to *Classical and Quantum Gravity* (CQG); currently under review. Preprint: arXiv:2605.11734 [astro-ph.HE].

Introduces mirror-paired relativistic time-delay asymmetries in Kerr spacetime as a potential observable signature of frame dragging and black-hole spin.

Presentations

Contributed Talk: 33rd Texas Symposium on Relativistic Astrophysics, Arizona State University (Dec 8–12, 2025) — *Kerr Polarization Transport: Accuracy and Performance in General Relativistic Light Propagation.*

Poster Presentation: CCAPP Poster Presentation, City College of New York (Dec 9, 2025) — *Kerr Polarization Transport: Accuracy and Performance in General Relativistic Light Propagation.*

Scholarships & Awards

Dean's List — Stony Brook University (Fall 2016, Spring 2020)
Dean's List — City College of New York (Spring 2024, Fall 2024, Spring 2025, Fall 2026)

Technical Skills

Programming & Scientific Computing: Python (NumPy, SciPy, Pandas, Matplotlib), C++, Fortran, Git, Jupyter

Scientific Writing & Data Analysis: LaTeX/Overleaf, IDL, ADQL/Gaia

Astrophysics & Computational Methods: Relativistic ray tracing, polarization transport, numerical integration methods, Gaia photometric calibration, full-Stokes imaging

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References

Prof. James M. Lattimer
Stony Brook University, Department of Physics and Astronomy
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Prof. Timothy H. Boyer
City College of New York, Department of Physics
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Prof. Joshua Tan
LaGuardia Community College (CUNY); Research Associate, AMNH
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Prof. Seth Cottrell
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